A Project Report

on

"Food Trends Understanding Customer Preferences in Food & Beverages"

Prepared by

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Submitted to

Infosys

INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP 6.0

Internship Duration:

February 2026 - March 2026

Certificate

This is to certify that **Ms. Pooja More**, a student of Computer Science and Engineering at MGM's College of Engineering, Nanded (M.S.), has successfully completed the project titled **"Food Trends Understanding Customer Preferences in Food & Beverages"** as part of the Infosys Springboard Virtual Internship 6.0 program.

The project work was carried out during the internship period from February 2026 to March 2026. During this internship, the student worked on analyzing food industry datasets to understand customer preferences and generate meaningful insights using data analytics and visualization techniques. This project report is submitted in partial fulfillment of the requirements of the Infosys Springboard Virtual Internship Program.

We wish her success in all her future academic and professional endeavors.

Mrs. Kalaiyarasi Gomathi
Project Mentor

ACKNOWLEDGMENT

I would like to express my sincere gratitude to Infosys Springboard for providing me with the opportunity to participate in the Virtual Internship 6.0 program and to work on the project titled **"Food Trends Understanding Customer Preferences in Food & Beverages."** This internship provided valuable exposure to data analytics and visualization techniques and helped me gain practical insights into analyzing real-world datasets.

I would like to extend my heartfelt thanks to my project mentor, **Mrs. Kalaiyarasi Gomathi** for her continuous guidance, valuable suggestions, and support throughout the completion of this project. Her mentorship played a significant role in the successful completion of this work.

I also express my sincere appreciation to the Department of Computer Science and Engineering, MGM's College of Engineering, Nanded, for their support and encouragement during the internship period.

With Deep Reverence,

Ms. Pooja More

ABSTRACT

The food and beverage industry generates large amounts of data related to customer preferences, product performance, and market trends. Analyzing this data helps businesses understand customer behavior and make informed decisions. The objective of this project, titled “Food Trends Understanding Customer Preferences in Food & Beverages,” was to analyze food industry data and present meaningful insights through interactive dashboards and visual reports.

In this project, data analysis and visualization techniques were used to examine customer preferences, product performance, and market trends. Multiple dashboards were created to present insights related to customer segmentation, product insights, sales trends, and strategy and innovation metrics. These dashboards helped in identifying patterns in customer behavior, popular product categories, and key factors influencing purchasing decisions.

The project also involved preparing a presentation to clearly communicate the insights obtained from the dashboards. Through visual analytics, the dashboards provide an easy and effective way to interpret complex datasets and support data-driven decision making in the food and beverage industry.

Overall, this project demonstrates how data visualization and dashboard analysis can help organizations understand customer trends, improve product strategies, and enhance business decision-making processes.

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INTRODUCTION

The food and beverage industry is one of the most dynamic sectors, where understanding customer preferences and market trends plays a crucial role in business success. With the rapid growth of digital platforms and data collection systems, organizations now have access to large volumes of data related to consumer behavior, product demand, and purchasing patterns. Analyzing this data effectively helps businesses identify trends, improve products, and make strategic decisions.

Data analytics and visualization techniques allow organizations to convert raw data into meaningful insights. Dashboards and visual reports are widely used tools that help decision-makers understand complex datasets quickly and efficiently. By presenting information through charts, graphs, and interactive visual elements, dashboards enable businesses to monitor performance, identify opportunities, and respond to changing customer preferences.

The project titled "Food Trends Understanding Customer Preferences in Food & Beverages" focuses on analyzing food industry data to gain insights into customer behavior and product performance. In this project, multiple dashboards were developed to visualize key aspects such as customer segmentation, product insights, and strategy and innovation metrics. These dashboards help in identifying popular product categories, customer demographics, and factors influencing purchasing decisions.

Additionally, a detailed presentation was prepared to communicate the insights derived from the dashboards. The combination of data analysis, visualization, and presentation helps in providing a clear understanding of market trends and customer preferences in the food and beverage sector.

This project demonstrates how data-driven insights and visual analytics can support organizations in making informed decisions, improving product strategies, and enhancing overall business performance.

PROBLEM STATEMENT

In today's digital era, the food and restaurant industry generates a vast amount of data through customer reviews, ratings, product feedback, and online ordering platforms. This data contains valuable information about customer preferences, popular cuisines, pricing patterns, and emerging food trends. However, many businesses face challenges in effectively analyzing and interpreting this data to support strategic decision-making. Without proper analysis, organizations may struggle to identify customer needs, optimize product offerings, and improve service quality.

The lack of structured insights from raw data can lead to missed opportunities for growth and reduced customer satisfaction. Therefore, there is a need for a data-driven approach that can transform raw restaurant and food trend data into meaningful insights. By using analytical tools and visualization techniques, businesses can better understand consumer behavior, identify high-demand food categories, and make informed decisions. This project aims to analyze food-related data and present key insights through dashboards and visualizations to support better business strategies and enhance customer experience.

Furthermore, with the rapid growth of digital platforms and food delivery services, competition in the food industry has increased significantly. Businesses must continuously monitor changing customer preferences and market trends to remain competitive. Traditional methods of analyzing business performance are often time-consuming and may not provide clear insights from large datasets. By applying data analysis techniques and visual dashboard tools, organizations can quickly interpret complex data and identify patterns that support better planning and decision-making. This project therefore focuses on transforming raw food trend data into meaningful visual insights that can assist businesses in understanding market behavior and improving their overall strategies.

PROJECT OBJECTIVES

The main objective of this project is to analyze food industry data and extract meaningful insights related to customer preferences, product performance, and market trends using data visualization techniques. The project focuses on presenting the analyzed data through interactive dashboards and clear visual reports to support better understanding and decision-making.

The specific objectives of the project are as follows:

- To analyze customer data in order to understand different customer segments and their preferences in the food and beverage industry.
- To identify product trends and performance by examining sales patterns, popular product categories, and customer demand.
- To create interactive dashboards that present complex data in a simple and visually understandable format.
- To study customer behavior and purchasing patterns using data analysis techniques.
- To generate meaningful insights that can help businesses make data-driven decisions and improve their product strategies.
- To present the insights effectively through a well-structured presentation and dashboard visualizations.

Through these objectives, the project aims to demonstrate how data analytics and visualization tools can help organizations better understand market trends and customer preferences in the food and beverage sector.

METHODOLOGY

The methodology of this project focuses on analyzing food industry data and presenting meaningful insights using data visualization techniques. A structured approach was followed to convert raw data into interactive dashboards and analytical insights.

1. Data Collection: Relevant datasets related to the food and beverage industry were used for the analysis. The dataset included information about customer preferences, product categories, sales trends, and purchasing behavior.

2. Data Preparation: The collected data was organized and prepared for analysis. This step involved reviewing the dataset, identifying inconsistencies, and structuring the data properly to ensure accurate visualization and interpretation.

3. Data Analysis: The prepared data was analyzed to identify patterns and relationships between different variables. This helped in understanding customer behavior, popular food categories, and emerging trends in the market.

4. Dashboard Development: Based on the analysis, interactive dashboards were created to visually represent the insights. The dashboards included various charts and graphs to highlight important aspects such as customer segmentation, product performance, and strategic trends.

5. Insight Presentation: The final insights derived from the dashboards were summarized and presented through a structured presentation. This helped in clearly communicating the findings and demonstrating how data visualization can support better decision-making.

TOOLS & TECHNOLOGIES

The project was developed using various tools and technologies to perform data analysis, cleaning, and visualization. These tools helped in transforming raw data into meaningful insights through interactive dashboards.

Power BI Desktop was used as the primary tool for dashboard creation and data visualization. It enabled the development of interactive dashboards to analyze different aspects such as customer segmentation, product insights, sentiment analysis, and strategy trends.

Power Query Editor was used for data cleaning and transformation. It helped in preparing the dataset by removing inconsistencies, handling missing values, and organizing the data into a structured format suitable for analysis.

DAX (Data Analysis Expressions) was used to create calculated columns and measures required for performing advanced data analysis. These calculations helped in deriving important metrics and improving the analytical capabilities of the dashboards.

The project dataset was stored in **Microsoft Excel / CSV format**, which served as the data source for Power BI. The dataset used in this project was obtained from a Swiggy Restaurant dataset available on Kaggle, which contains information related to restaurant details, ratings, and customer preferences.

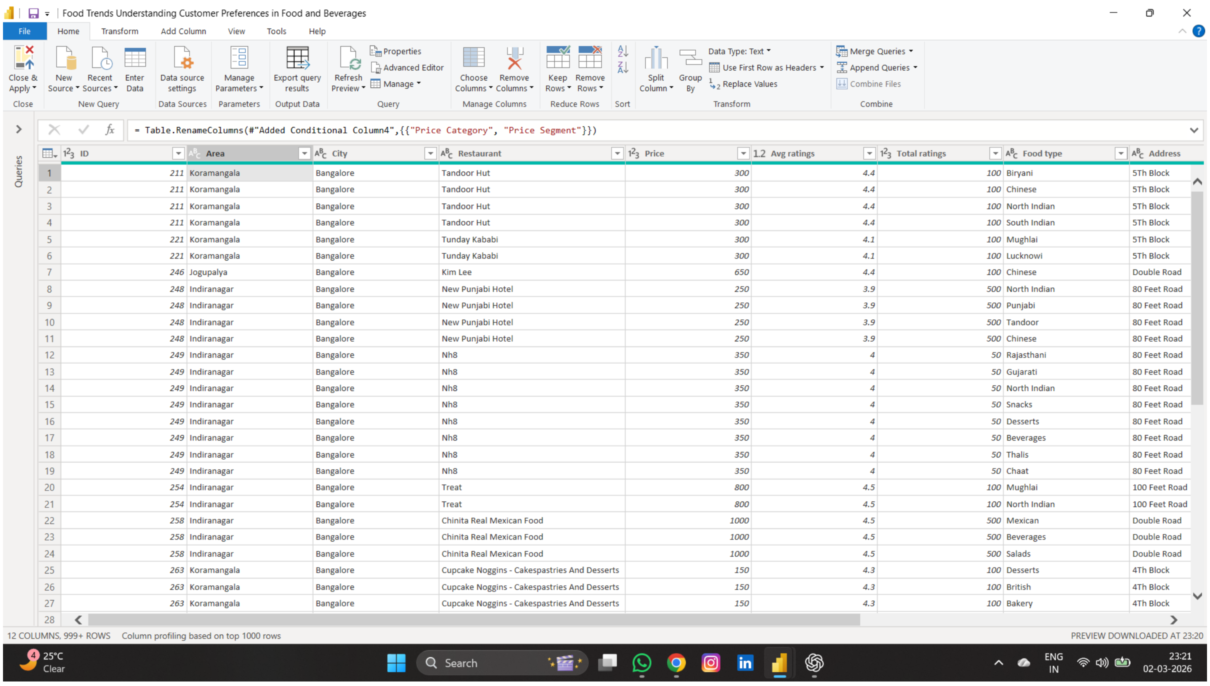
Overall, this project demonstrates the practical application of Business Intelligence and Data Analytics concepts by using modern data visualization tools to analyze and present insights from real-world datasets.

DATA PREPARATION & CLEANING

Before performing the analysis and creating dashboards, the dataset was cleaned and prepared to ensure accuracy and consistency. Data cleaning is an essential step in data analytics because raw datasets often contain missing values, duplicate entries, or inconsistent formatting.

The dataset used in this project was obtained from a Swiggy Restaurant dataset available on Kaggle, which includes information related to restaurants, food categories, ratings, customer reviews, and pricing details. After importing the dataset into Power BI, the Power Query Editor was used to perform data cleaning and transformation tasks.

During the preprocessing stage, duplicate records were removed, missing values were handled, and inconsistent data formats were corrected. The dataset was then organized into a structured format suitable for analysis. Unnecessary columns were removed, while important attributes such as restaurant name, cuisine type, rating, price range, and customer feedback were retained for further analysis.



The cleaned dataset provided a reliable foundation for building dashboards and generating insights related to customer preferences and product trends in the food and beverage industry.

EXECUTIVE OVERVIEW

The Executive Overview Dashboard provides a comprehensive summary of the key insights derived from the dataset. This dashboard serves as the starting point for understanding the overall trends and patterns present in the food and beverage industry data. It is designed to present important information in a simple and visually structured format so that users can quickly grasp the most significant aspects of the dataset.

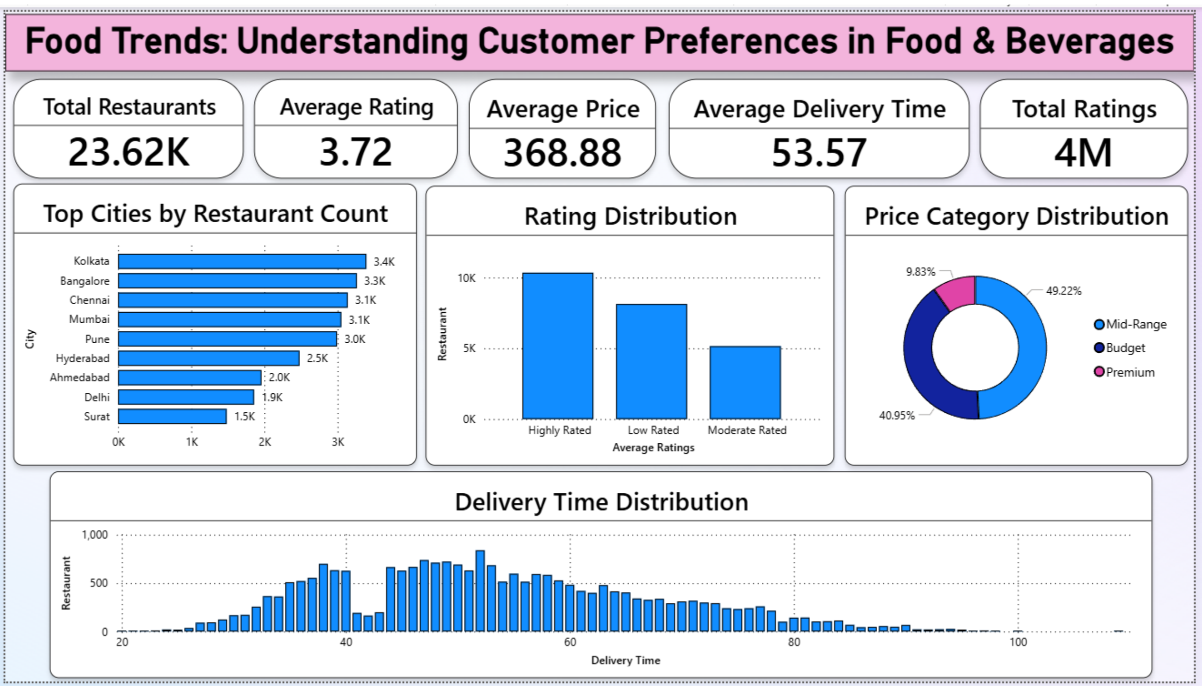
The dashboard includes several key indicators that highlight important metrics such as the total number of restaurants analyzed, the average customer rating, pricing patterns, and the distribution of various food categories. These metrics help in providing a quick overview of the market and offer a general understanding of restaurant performance and customer preferences.

Various visual elements such as bar charts, pie charts, and summary cards are used in the dashboard to represent the data effectively. These visualizations make it easier to identify patterns and trends within the dataset. For example, the charts help in identifying the most popular cuisines, the distribution of restaurants across different categories, and the variation in customer ratings across different food outlets. Such visual insights allow users to quickly interpret complex data without the need for extensive numerical analysis.

The Executive Overview Dashboard also helps in highlighting key trends that can influence business decisions. By providing a consolidated view of important information, the dashboard enables stakeholders to identify areas of high demand, understand customer preferences, and evaluate the overall performance of different restaurant categories.

This dashboard acts as the foundation for deeper analysis in the project. After reviewing the high-level insights presented in this dashboard, users can further explore detailed dashboards such as Sentiment Analysis, Product Insights, Customer Segmentation, and Strategy and Innovation dashboards. These dashboards provide more specific insights into different aspects of the food and beverage market.

Overall, the Executive Overview Dashboard plays an important role in summarizing the dataset and presenting key information in a clear and interactive format. It helps in simplifying complex datasets and supports effective data-driven decision-making by providing a quick and informative overview of the market trends.



SENTIMENT ANALYSIS

The Sentiment Analysis Dashboard focuses on analyzing customer opinions and feedback related to restaurants and food services. Customer reviews and ratings play an important role in understanding customer satisfaction and overall service quality in the food and beverage industry. This dashboard was developed to examine the overall sentiment of customers and identify patterns in their feedback.

The dashboard categorizes customer feedback into different sentiment categories such as positive, neutral, and negative. By analyzing these sentiments, it becomes easier to understand how customers perceive different restaurants and food services. A higher number of positive sentiments generally indicates higher customer satisfaction, while negative sentiments may highlight areas where improvements are needed.

Various visualizations such as bar charts, pie charts, and rating distribution graphs were used to represent the sentiment patterns in a clear and understandable way. These visual elements allow users to quickly observe how customers rate their experiences and how sentiment varies across different restaurants or food categories.

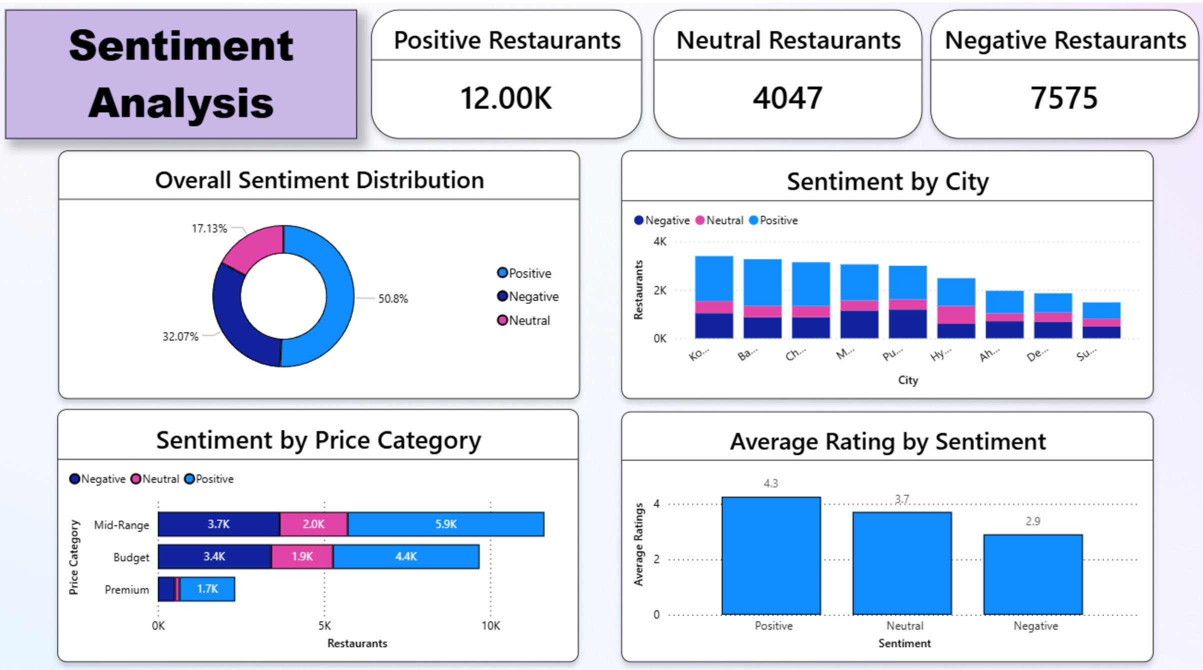
The dashboard also helps in identifying trends related to customer satisfaction. For example, restaurants with higher ratings often show a greater proportion of positive feedback, while restaurants with lower ratings may receive more negative responses. Such insights help businesses understand the factors influencing customer satisfaction and identify opportunities for improvement.

In addition, the Sentiment Analysis Dashboard supports decision-making by highlighting customer expectations and preferences. Businesses can use these insights to improve service quality, enhance food offerings, and address customer concerns more effectively.

Overall, the Sentiment Analysis Dashboard provides valuable insights into customer opinions and helps businesses understand how customers perceive their services. By analyzing sentiment trends, organizations can make informed decisions to improve customer satisfaction and strengthen their position in the competitive food and beverage market.

Furthermore, sentiment analysis also helps organizations monitor customer perception over time. By continuously analyzing feedback and reviews, businesses can track changes in customer satisfaction levels and identify emerging trends in customer preferences. This enables companies to respond quickly to customer needs and maintain a competitive advantage in the market.

In addition, the insights obtained from the Sentiment Analysis Dashboard can support strategic planning and service improvement initiatives. Understanding customer sentiment allows businesses to focus on areas that require attention, improve operational efficiency, and deliver better customer experiences. As a result, sentiment analysis becomes an important tool for enhancing customer loyalty and ensuring long-term business growth.



PRODUCT INSIGHTS

The Product Insights Dashboard focuses on analyzing the performance and popularity of different food products and categories available in the dataset. Understanding product trends is important for businesses in the food and beverage industry because it helps them identify which food categories are most preferred by customers and which products perform well in the market.

This dashboard provides a detailed view of product-related information such as the distribution of various food categories, restaurant offerings, pricing patterns, and customer ratings associated with different products. By analyzing these factors, businesses can gain valuable insights into customer demand and product performance.

Several visualization techniques such as bar charts, pie charts, and comparative graphs were used in this dashboard to present product-related insights clearly. These visualizations help in identifying the most popular cuisines, the categories with the highest number of restaurants, and the products that receive the highest customer ratings. Such information is useful for understanding market demand and identifying trending food items.

The Product Insights Dashboard also highlights the relationship between product categories and customer satisfaction. By analyzing ratings and reviews associated with different food categories, it becomes possible to determine which products are most appreciated by customers. Restaurants offering highly rated products are likely to attract more customers and maintain stronger customer loyalty.

In addition, the dashboard helps in identifying product diversity and variety within the dataset. It shows how restaurants differentiate themselves by offering different cuisines and menu options. This insight is important for understanding competitive positioning within the food and beverage market.

Furthermore, the dashboard enables stakeholders to evaluate product performance across different segments and categories. By observing patterns in ratings, pricing, and product availability, businesses can identify which food items are consistently performing well and which ones may require improvement or repositioning in the market.

Another important aspect highlighted by the dashboard is the potential opportunity for introducing new products or improving existing offerings. By understanding customer preferences and identifying gaps in the market, businesses can develop innovative menu items and enhance their product strategies to better meet customer expectations.

Overall, the Product Insights Dashboard provides valuable information about product performance and customer preferences. The insights obtained from this dashboard can help businesses improve their product offerings, focus on popular food categories, and design better menu strategies to meet customer demand.

CUSTOMER SEGMENTATION

The Customer Segmentation Dashboard focuses on analyzing different groups of customers based on their preferences, behavior, and interaction with restaurants in the dataset. Customer segmentation is an important concept in data analytics because it helps businesses understand their customers more effectively and design strategies that cater to different customer groups.

This dashboard provides insights into how customers interact with various restaurants and food categories. By examining factors such as ratings, pricing levels, and cuisine preferences, it becomes possible to identify patterns in customer behavior. Understanding these patterns allows businesses to recognize which types of customers prefer certain cuisines or restaurant categories.

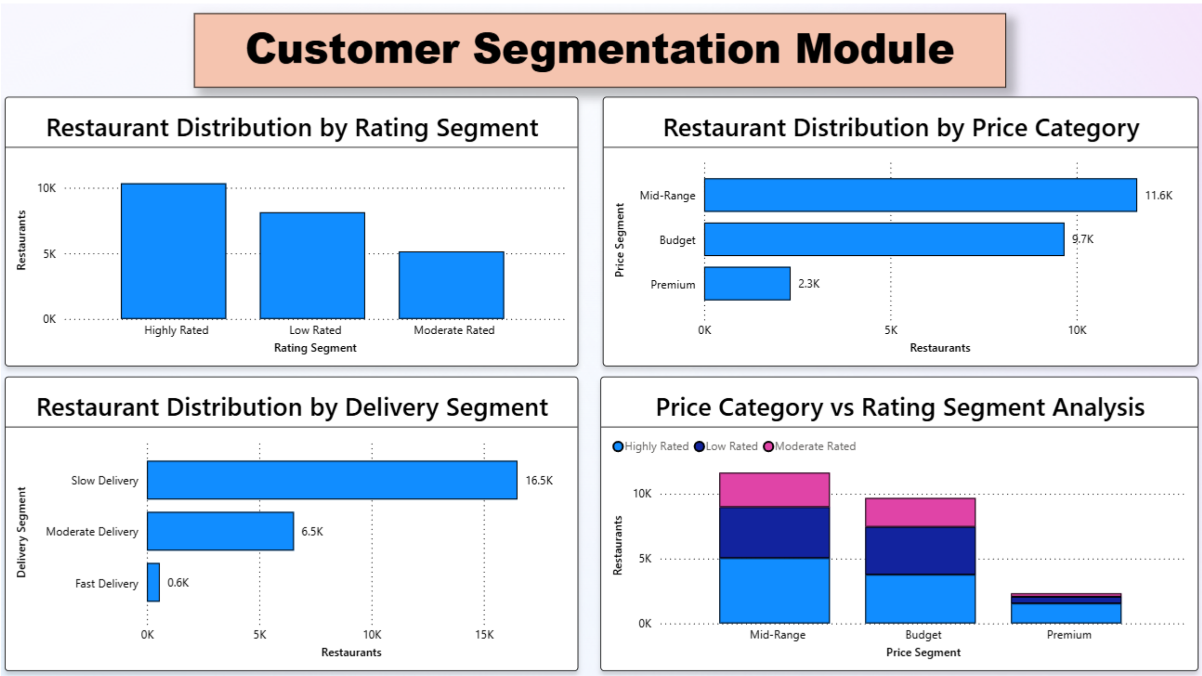
Several visualizations such as bar charts, category comparisons, and distribution charts were used in the dashboard to represent customer patterns clearly. These visual elements make it easier to observe how customer preferences vary across different segments. For example, some customers may prefer premium restaurants with higher pricing levels, while others may prefer affordable food options with moderate ratings.

The dashboard also highlights how different food categories attract different groups of customers. Certain cuisines may receive higher ratings from specific customer segments, indicating strong customer interest in those categories. Such insights help restaurants understand their target audience and tailor their offerings according to customer expectations.

Another important aspect of the Customer Segmentation Dashboard is identifying trends related to customer satisfaction. By analyzing rating distributions and customer preferences, it becomes easier to determine which types of restaurants are most successful in meeting customer expectations. Restaurants that consistently receive higher ratings are likely to have better service quality, food taste, and overall customer experience.

Additionally, the insights derived from customer segmentation can help businesses design more effective marketing strategies. By understanding customer groups and their preferences, businesses can create targeted promotions, introduce new menu items that align with customer demand, and improve overall service quality. This approach enables organizations to strengthen customer engagement and build long-term customer relationships.

Overall, the Customer Segmentation Dashboard plays a crucial role in understanding customer behavior and identifying different customer groups within the dataset. The insights obtained from this dashboard help businesses make informed decisions regarding product offerings, pricing strategies, and customer engagement initiatives.



STRATEGY & INNOVATION

The Strategy and Innovation Dashboard focuses on identifying strategic insights and opportunities for improvement within the food and beverage industry. This dashboard helps in analyzing important business factors such as restaurant performance, customer preferences, and market trends. By studying these aspects, organizations can develop effective strategies to improve their services and maintain a competitive position in the market.

The dashboard presents various visualizations that highlight key indicators such as restaurant ratings, pricing patterns, and the distribution of different food categories. These visual elements allow users to observe trends and understand how different factors influence customer satisfaction and restaurant performance. Such insights help businesses identify areas where improvements or innovations can be introduced.

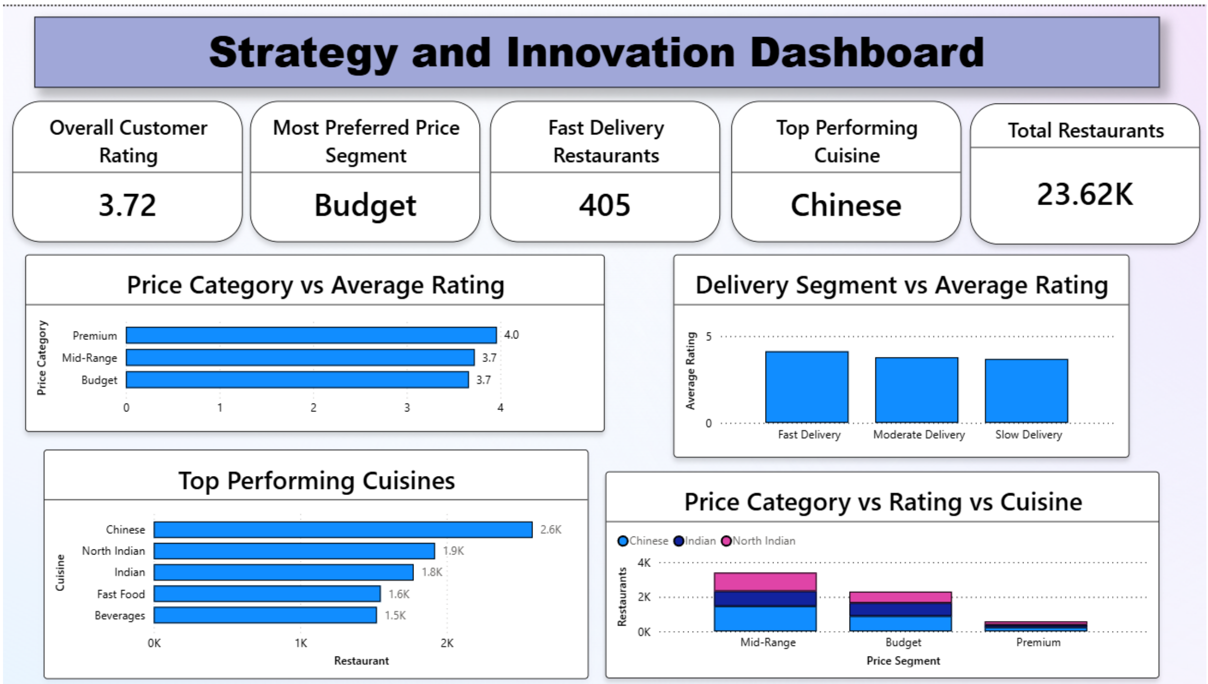
One of the important objectives of this dashboard is to support strategic decision-making. By analyzing the performance of different restaurants and cuisines, businesses can determine which areas are performing well and which require improvement. For example, if certain cuisines receive consistently high ratings, restaurants may focus on expanding those menu options to attract more customers.

The dashboard also helps identify opportunities for innovation in the food and beverage sector. Businesses can explore new food categories, improve menu offerings, or introduce new service strategies based on the insights obtained from the analysis. Innovation plays a significant role in attracting customers and maintaining long-term business growth.

In addition, the Strategy and Innovation Dashboard helps businesses understand market competition and customer expectations. By analyzing the relationship between pricing, ratings, and food categories, organizations can adjust their pricing strategies and service quality to better match customer needs. This allows businesses to remain competitive while delivering better value to customers.

Furthermore, the insights obtained from this dashboard can guide long-term business planning. Companies can use these findings to develop strategies that improve customer satisfaction, expand product offerings, and enhance overall business performance. The ability to analyze data and identify strategic opportunities enables organizations to make informed decisions and adapt to changing market conditions.

Overall, the Strategy and Innovation Dashboard provides valuable strategic insights that support business growth and innovation in the food and beverage industry. By combining data analysis with visual dashboards, businesses can better understand market trends and implement effective strategies for future success.



KEY INSIGHTS

The analysis of the dataset through multiple dashboards provided valuable insights into customer preferences, restaurant performance, and trends within the food and beverage industry. The dashboards helped convert complex data into clear visual information, making it easier to identify patterns and understand customer behavior.

One of the key findings from the analysis is that customer preferences vary significantly across different food categories and cuisines. Certain cuisines appear more popular among customers, indicating higher demand in the market. Restaurants offering these popular cuisines generally receive better ratings and attract a larger number of customers.

The sentiment analysis also highlighted important patterns in customer satisfaction. Restaurants with higher ratings tend to receive more positive feedback, suggesting that food quality, service efficiency, and overall customer experience play an important role in shaping customer opinions. In contrast, restaurants with lower ratings may need to improve their service quality and customer experience to enhance their reputation.

Product insights revealed that some food categories consistently perform better than others in terms of customer ratings and popularity. These categories represent potential opportunities for restaurants to expand their menu offerings and attract more customers.

The Customer Segmentation Dashboard further helped identify how different groups of customers interact with restaurants and food categories. Different customer segments often prefer different cuisines, pricing levels, and restaurant types. Understanding these segments can help businesses design targeted strategies and improve customer engagement.

Overall, the insights obtained from the dashboards demonstrate how data analytics and visualization can help businesses understand market trends, improve customer satisfaction, and make more informed strategic decisions.

BUSINESS RECOMMENDATIONS

Based on the insights obtained from the dashboard analysis, several recommendations can be made to improve business performance in the food and beverage industry. These recommendations focus on enhancing customer satisfaction, improving product offerings, and supporting data-driven decision-making.

One important recommendation is that restaurants should maintain high standards of food quality and customer service. Customer ratings and reviews indicate that the overall dining experience strongly influences customer satisfaction. Restaurants that consistently provide high-quality food and efficient service are more likely to receive positive feedback and attract repeat customers.

Another recommendation is to focus on the most popular cuisines and food categories identified through the analysis. Restaurants can expand their menu options within these high-demand categories to attract a larger customer base. Introducing innovative menu items or seasonal offerings can also help businesses differentiate themselves from competitors.

Businesses should also use customer feedback and sentiment analysis to monitor customer opinions and expectations. Analyzing customer reviews helps identify areas where improvements are required, such as service quality, pricing, or menu variety. Addressing these issues can significantly enhance the overall customer experience.

Finally, restaurants should adopt data analytics and business intelligence tools to support decision-making. By using dashboards and analytical tools, businesses can track performance indicators, identify trends, and make informed decisions based on reliable data. A data-driven approach enables restaurants to improve operational efficiency and remain competitive in the market.

FUTURE SCOPE

Although this project successfully analyzes customer preferences and food industry trends using data visualization techniques, there are several opportunities for further improvement and expansion in the future. Enhancing the scope of the analysis can provide deeper insights and support more advanced decision-making in the food and beverage industry.

One possible improvement is the use of larger and more diverse datasets. The current analysis is based on a specific restaurant dataset, but future studies could integrate data from multiple food delivery platforms or restaurant chains. This would allow a more comprehensive understanding of customer preferences, regional food trends, and variations in customer behavior across different locations.

Another potential area of development is the use of advanced analytics techniques such as machine learning and predictive modeling. These techniques can help predict future customer preferences, forecast food demand, and identify emerging trends in the market. Predictive models could assist businesses in planning their menu offerings and improving operational strategies.

Future work can also include more advanced sentiment analysis using Natural Language Processing (NLP). Instead of relying only on rating values, text-based customer reviews can be analyzed to understand detailed opinions related to food quality, service experience, pricing, and delivery efficiency.

Overall, expanding the dataset and incorporating advanced analytical methods can significantly enhance the capabilities of the project and help businesses make more effective data-driven decisions in the food and beverage industry.

CONCLUSION

The project titled "Food Trends Understanding Customer Preferences in Food & Beverages" demonstrates how data analytics and visualization techniques can be used to analyze customer behavior and identify market trends in the food and beverage industry. By using business intelligence tools and structured datasets, meaningful insights were derived from the available restaurant and customer data.

Through the development of multiple dashboards such as the Executive Overview, Sentiment Analysis, Product Insights, Customer Segmentation, and Strategy and Innovation dashboards, the project provided a comprehensive understanding of restaurant performance and customer preferences. These dashboards helped transform complex datasets into clear visual insights, making it easier to identify patterns, trends, and opportunities within the market.

The analysis highlighted the importance of customer satisfaction, product quality, and strategic decision-making in improving business performance. Restaurants that understand customer preferences and continuously analyze market trends are better positioned to enhance their services and attract more customers.

Overall, this project successfully demonstrates the practical application of data analytics and business intelligence concepts using modern visualization tools. The insights obtained from the analysis can support businesses in making informed decisions and developing effective strategies to improve their products, services, and customer experience in the competitive food and beverage industry.